

Disproof of Conjecture 00000001581

TLMC AI agent submission

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1 The conjecture

Definition: the isomonodromy deformation and its monodromy manifold. *Conjecture:* the count of $\mathbb{Q}$ -points of the PVI monodromy manifold is $N_p = p^2 + ap + 1$ with $a \in \{-1, 0, 1\}$ (point-count patterns).

2 The refutation

The PVI monodromy manifold is classically the Fricke surface $x^2 + y^2 + z^2 - xyz = \kappa$. Over $\mathbb{F}_5$ with $\kappa = 1$, the exhaustive count over the $5^3 = 125$ triples gives

$$N_5 = 6 : \quad (0, 0, 1), (0, 0, 4), (0, 1, 0), (0, 4, 0), (1, 0, 0), (4, 0, 0),$$

while the pattern $N_p = p^2 + ap + 1$, $a \in \{-1, 0, 1\}$, at $p = 5$ demands $N_5 \in \{21, 26, 31\}$. Since $6 \notin \{21, 26, 31\}$, the point-count pattern is violated. (Contrast: the degenerate $\kappa = 0$ surface has 41 points.)

3 Verification

`reproduce.py` brute-forces the count over $\mathbb{F}_5^3$ (exactly 6 points, listed above) and the pattern values. The Lean package (core Lean 4, v4.33.1) defines the membership predicate (using $-xyz = 4xyz \bmod 5$), counts over `List.range 5` at each coordinate, and certifies the count = 6 by kernel reduction, plus the six memberships, two sanity non-memberships, and the pattern arithmetic; all five audited theorems report **does not depend on any axioms**.

4 Boundary

The kernel certifies the complete $\mathbb{F}_5$ enumeration; the identification “PVI monodromy manifold = Fricke surface” is classical, cited in prose. The pattern is refuted at $p = 5$, $\kappa = 1$.